

MOUNDHIR BACHENE

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building mobile apps and SaaS products that businesses rely on every day. I am a problem-solver first—my focus is on fixing the core issue and delivering a working product. From launching multi-role ecosystems like food delivery and artisan marketplaces to building resilient enterprise tools, I adapt quickly, use modern AI tools to accelerate development, and take complex systems from concept to production.

PROFESSIONAL EXPERIENCE

Founder & Lead Software Engineer | Moody – Food & Mood | Food Ordering + Restaurant Management + Driver Dispatch & Delivery Tracking | Jan 2025 – Present

Leading product design, development and launch for a two-sided food app (consumer, restaurant & driver), for Gen Z users & local restaurants.

- Built the full app using Flutter, Riverpod and Firebase (Auth, Realtime DB, FCM, security rules).
- Merchant features: order lifecycle (accept → ready → complete), menu/prices, stock, discounts & promo codes.
- Consumer features: multi-restaurant cart, delivery/pickup, location-based filtering and real-time updates.
- Handled the full product lifecycle: roadmap, QA, deployment strategy, and user onboarding for launch.
- Implemented guest mode with guest-to-user data transfer and strong error and permission handling.
- Improved performance (lazy load, image caching, debounced search) and applied clean architecture patterns.
- Driver app (3rd layer): full-screen order offer with 60s countdown, Firebase first-accept-wins acceptance, and automatic broadcast to nearby drivers using Haversine distance filtering.
- Streamed real-time GPS to the consumer app during active deliveries, with live tracking on Google Maps and a 4-phase state machine (heading → arrived → picked up → delivered) triggering order status updates via Cloud Functions.
- Applied Algerian data protection Law 18-07: driver identity data stored in device-local secure storage only.

Software Engineer | Artisan Marketplace App | April 2026 – Present

Two-sided marketplace connecting clients with local artisans across Algeria. Single Flutter app with two user roles (client + artisan).

- Built the full booking flow: artisan discovery, contact unlock, quote exchange, and in-app credit payment.
- Implemented a wallet system via Cloud Functions (TypeScript): free credits on signup, Firestore transaction-based transfers, and full transaction history.
- Artisan onboarding: phone OTP, document upload, admin approval queue, profile setup, and portfolio management.
- Real-time chat, FCM push notifications with deep-link routing and live booking status and balance streams.
- Stack: Flutter, Riverpod, GoRouter, Clean Architecture, Firebase (Firestore, Auth, Storage, Cloud Functions, FCM), Flutter Web admin dashboard.

Software Engineer | Van Distribution (Offline-First) | Nov 2025 – Jan 2026

Direct Store Delivery (DSD) and Van Sales mobile solution for distributors and wholesalers that helps field teams make sales, track inventory across depots, and close daily accounts. While operating entirely offline — reducing manual entry, reducing stock errors, and ensuring accurate matching with central systems.

- Built fast mobile DSD works offline using Flutter and SQLite to support full business operations.
- Implemented safe updates and sync that avoids conflicts to ensure financial and stock accuracy.

- Built XML-based enterprise synchronization (DATAPACKET format): morning import/evening export
- Integrated Bluetooth printing, PDF reporting, barcode scanning, and GPS geo-tagging.
- Developed multi-depot inventory logic (unit/carton tracking) and ledger-style financial modules (CA, payments, balances).
- Implemented device-based activation and licensing control.

Software Engineer | Stock Manager | Aug 2025 – Nov 2025

Professional inventory and sales manager for retail/wholesale stores that enforces data accuracy, automates credit/payment reconciliation, and secures local access — helping reduce inventory errors, speed up cash closing, and protect device access.

- Developed an offline-first inventory system using Flutter and SQLite with secure local persistence.
- Designed relational database schemas for inventory, suppliers, clients, and debt tracking.
- Implemented credit tracking with automatic payment allocation across outstanding balances and advances.
- Built daily closing (Clôture) workflows to track daily turnover and profit margins, capturing costPrice at time of sale.
- Developed secure login (activation keys, PIN, biometrics); Integrated PDF receipts and Bluetooth thermal printing.

Software Engineer | Van Distribution (Cloud-Sync) | Oct 2024 – Mar 2025

Mobile Distribution Management System that lets distributors (multi-van field teams) perform sales, manage inventory and reconcile finances in low- or no-connectivity environments — removing manual paperwork, preventing stock errors, and speeding up daily financial closing.

- Built DSD system using Flutter with BLoC and Repository Pattern.
- Implemented hybrid storage architecture (SQLite + Firebase Realtime DB) with sync that updates when connected.
- Designed custom Firebase caching engine (TTL + memory optimization) for low-connectivity environments.
- Developed safe money update logic ensuring accurate turnover, payments, and balance updates.
- Integrated dynamic PDF generation and Bluetooth thermal invoice printing.
- Built the mobile app to work with the desktop system in real time.

Blockchain & Data Analyst | Krea8te (Sastanaqqam) | Algiers, Dubai | Apr 2022 – Sep 2024

Technical Execution & QA: Managed MPC wallet configuration, executed token vesting contracts, and led QA for NFT marketplace features (minting, rentals, royalties, WalletConnect).

Product & Documentation: Designed wallet UX flows and collaborated on whitepapers and investor pitch decks.

DeFi Analysis: Analyzed DeFi protocols and tokenomics to identify DApp integration opportunities; represented the company at international blockchain events.

EDUCATION

MSc in Fundamental Informatics & Information Technology | People's Friendship University of Russia (RUDN) Moscow | 2019 – 2021
BSc in Computer Science | Yahia Fares University of Medea — Medea, Algeria | 2015 – 2018

TECHNICAL SKILLS

- **Frameworks & Languages:** Flutter, Dart, Material 3, SQL (SQLite).
- **State Management:** BLoC, Riverpod, Provider.
- **Architecture:** Clean Architecture, Repository Pattern, Offline-First Sync.
- **CI/CD:** GitHub Actions (flutter analyze, dart format, flutter test on every push).
- **Backend & Cloud:** Firebase (Firestore, Auth, Realtime DB, Cloud Functions, FCM).
- **Specialized Integrations:** Biometrics (local_auth), Bluetooth Thermal Printing, Barcode Scanning, GPS, PDF Generation.
- **Data & Analysis:** Blockchain & DeFi, Data Analysis, Tokenomics, Data Visualization.

LANGUAGES

Arabic: Native | English: Advanced (IELTS 7) | French: Advanced | Russian: Upper-Intermediate